

Square and Cube Roots

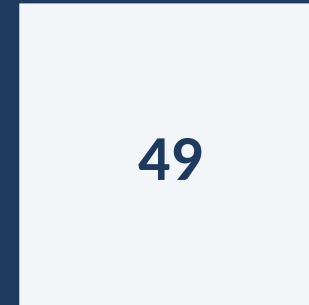
Lesson 01 • 1.5 • Class MYP 8A • 50 minutes

Objective

I can calculate and estimate square roots and cube roots.

Success criteria: I can...

- find $\sqrt{\quad}$ of a perfect square and $\sqrt[3]{\quad}$ of a perfect cube
- find $\sqrt[3]{\quad}$ of a negative number; explain why $\sqrt{\text{(negative)}}$ is not real
- say which two whole numbers a root lies between
- estimate a root to 1 d.p. and check it



$$\text{side} = \sqrt{49} = 7$$



$$\text{edge} = \sqrt[3]{8} = 2$$

Starter: squares and cubes

Worksheet page 1 · Section 1 · 3 minutes · no calculator

S1 $7^2 = ?$

S2 $11^2 = ?$

S3 $3^3 = ?$

S4 $10^3 = ?$

S5 $(-4)^2 = ?$

S6 Which is both a perfect square and a perfect cube? 16 27 64 100

Starter: check your answers

S1 $7^2 = 7 \times 7 = 49$

S2 $11^2 = 11 \times 11 = 121$

S3 $3^3 = 3 \times 3 \times 3 = 27$

S4 $10^3 = 10 \times 10 \times 10 = 1000$

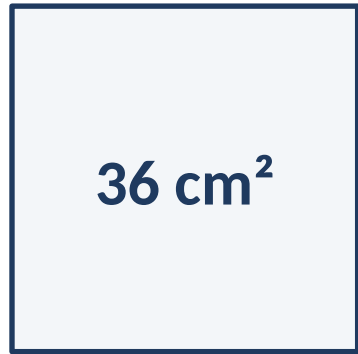
S5 $(-4)^2 = (-4) \times (-4) = 16$

S6 $64 = 8^2$ and $64 = 4^3$

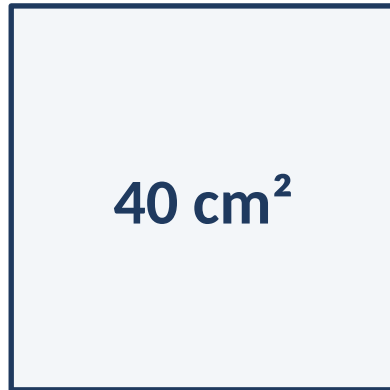
Watch out: 3^3 is NOT $3 \times 3 = 9$ • a negative number squared is positive

From your preparation: undoing squaring

Preparation question P9 · Worksheet page 1 · N1



side = ?



side = ?

Area 36 → side 6, because $6 \times 6 = 36$

Area 40 → side is not a whole number:

- $6 \times 6 = 36$ is too small
- $7 \times 7 = 49$ is too big
- so the side is between 6 and 7

Finding the side from the area undoes squaring. That is a square root.

Square roots and cube roots

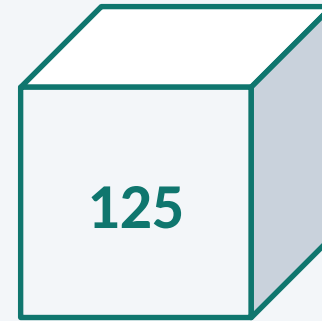


81

Square root $\sqrt{}$

$\sqrt{81} = 9$ because $9^2 = 81$

- $\sqrt{}$ undoes squaring
- $\sqrt{n}$ is never negative
- area of square $\rightarrow$ side



Cube root $\sqrt[3]{}$

$\sqrt[3]{125} = 5$ because $5^3 = 125$

- $\sqrt[3]{}$ undoes cubing
- $\sqrt[3]{}$ of a negative is negative
- volume of cube $\rightarrow$ edge

Example 1: calculate roots

Worksheet page 2 • Example 1 • annotate as I explain

(a) $\sqrt{81} = 9$

Which number $\times$ itself = 81? Check: $9^2 = 81$ ✓

(b) $\sqrt[3]{125} = 5$

Which number cubed = 125? Check: $5 \times 5 \times 5 = 125$ ✓

(c) $\sqrt[3]{-8} = -2$

Negative cubed is negative. Check: $(-2)^3 = -8$ ✓

(d) $\sqrt{-9}$: no real answer

$3^2 = 9$ and $(-3)^2 = 9$. No real number squares to -9 .

Always check: square or cube your answer.

Example 2: estimate $\sqrt{50}$ (trap it)

Worksheet page 2 • Example 2 • steps 1–3

1. Trap

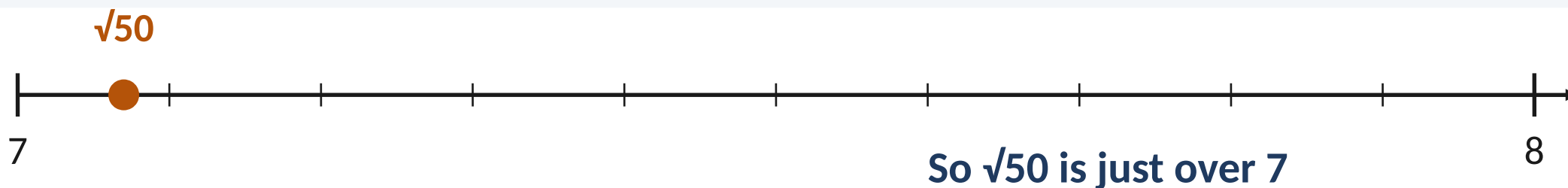
$49 < 50 < 64$ perfect squares either side of 50

2. Roots

$7 < \sqrt{50} < 8$ because $\sqrt{49} = 7$ and $\sqrt{64} = 8$

3. Closer?

50 is 1 away from 49 but 14 away from 64



Example 2: estimate $\sqrt{50}$ (test and decide)

Worksheet page 2 • Example 2 • steps 4–5

4. Test

$$7.0^2 = 49 \quad (\text{below } 50)$$

$$7.1^2 = 50.41 \quad (\text{above } 50) \rightarrow \sqrt{50} \text{ is between } 7.0 \text{ and } 7.1$$

5. Decide

50 is closer to 50.41 (0.41 away) than to 49 (1 away)

$$\sqrt{50} \approx 7.1 \quad (1 \text{ d.p.})$$

Example 3: estimate $\sqrt[3]{30}$

Worksheet page 2 • Example 3 • same five steps, with cubes

1. Trap

$27 < 30 < 64$ perfect cubes: $3^3 = 27$, $4^3 = 64$

2. Roots

$3 < \sqrt[3]{30} < 4$

3. Closer?

30 is 3 away from 27, 34 away from 64 $\rightarrow$ just over 3

4. Test

$3.1^3 = 29.791$ $3.2^3 = 32.768$

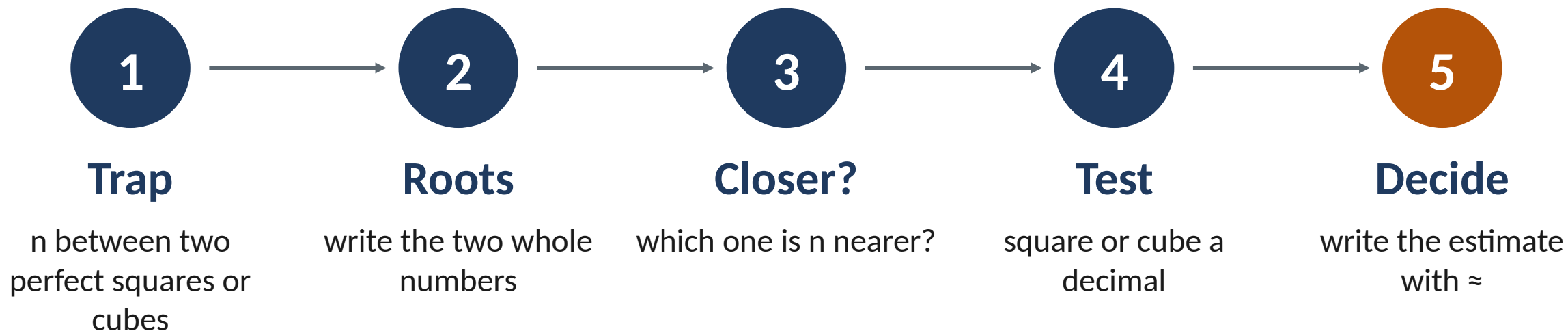
5. Decide

30 is much closer to 29.791 $\rightarrow \sqrt[3]{30} \approx 3.1$

Cube roots: use perfect CUBES, not perfect squares.

The method

Worksheet page 3 • Support Box



Sentence stem: " $\sqrt{\quad}$ is between $\quad$ and $\quad$ because $\quad < \quad < \quad$."

Round 1: calculate

Boards down → work alone → show in 3, 2, 1 → do not erase

1 $\sqrt{9}$

2 $\sqrt[3]{8}$

3 $\sqrt{64}$

4 $\sqrt[3]{-125}$

5 True or false? $\sqrt[3]{27} = 9$

Round 1: answers

1 $\sqrt{9} = 3$ check: $3^2 = 9$

2 $\sqrt[3]{8} = 2$ check: $2^3 = 8$

3 $\sqrt{64} = 8$ check: $8^2 = 64$ ($\sqrt[3]{64}$ is 4)

4 $\sqrt[3]{(-125)} = -5$ check: $(-5)^3 = -125$

5 FALSE: $\sqrt[3]{27} = 3$ check: $3^3 = 27$. $9^3 = 729$, not 27

W1: calculate

Worksheet page 3 · W1 · fill in the blanks

(a) $\sqrt{144}$ (b) $\sqrt[3]{64}$ (c) $\sqrt[3]{-27}$

(a) $\sqrt{144} = \underline{\quad}$ because $\underline{\quad}^2 = 144$

(b) $\sqrt[3]{64} = \underline{\quad}$ because $\underline{\quad}^3 = 64$

(c) $\sqrt[3]{-27} = \underline{\quad}$ because $(\underline{\quad})^3 = -27$

W1: solution

(a) $\sqrt{144} = 12$

$$12^2 = 12 \times 12 = 144 \checkmark$$

(b) $\sqrt[3]{64} = 4$

$$4^3 = 4 \times 4 \times 4 = 64 \checkmark \quad (\text{not } 8: 8^2 = 64)$$

(c) $\sqrt[3]{(-27)} = -3$

$$(-3)^3 = (-3) \times (-3) \times (-3) = -27 \checkmark$$

W2: estimate $\sqrt{20}$ to 1 d.p.

Worksheet page 3 · W2 · some steps started for you

Estimate $\sqrt{20}$ to 1 decimal place

Trap: $__ < 20 < __$ Roots: $__ < \sqrt{20} < __$

Closer to: $__$

Test: $4.4^2 = ______$ $4.5^2 = ______$

Decide: $\sqrt{20} \approx ______$

W2: solution

Trap

$$16 < 20 < 25$$

Roots

$$4 < \sqrt{20} < 5$$

Closer?

20 is 4 from 16, 5 from 25: near the middle, so test

Test

$$4.4^2 = 19.36 \quad 4.5^2 = 20.25$$

Decide

20 is 0.25 from 20.25 but 0.64 from 19.36 $\rightarrow \sqrt{20} \approx 4.5$

W3: estimate $\sqrt[3]{100}$ to 1 d.p.

Worksheet page 3 · W3 · no template: use the five steps

Estimate $\sqrt[3]{100}$ to 1 decimal place

Use the five steps: Trap → Roots → Closer? → Test → Decide

Perfect cubes: 1, 8, 27, 64, 125, 216 ...

W3: solution

Trap

$$64 < 100 < 125$$

Roots

$$4 < \sqrt[3]{100} < 5$$

Closer?

36 from 64, 25 from 125 $\rightarrow$ somewhere in the middle, test

Test

$$4.6^3 = 97.336 \quad 4.7^3 = 103.823$$

Decide

100 is closer to 97.336 (2.664 away) $\rightarrow \sqrt[3]{100} \approx 4.6$

Round 2: between and estimate

Boards down → work alone → show in 3, 2, 1 → do not erase

1 $\sqrt{10}$ is between ? and ?

2 $\sqrt{45}$ is between ? and ?

3 $\sqrt[3]{20}$ is between ? and ?

4 Estimate $\sqrt{30}$ to 1 d.p.

5 True or false? $\sqrt{-4} = -2$

Round 2: answers

1 $3 < \sqrt{10} < 4$ because $9 < 10 < 16$

2 $6 < \sqrt{45} < 7$ because $36 < 45 < 49$

3 $2 < \sqrt[3]{20} < 3$ because $8 < 20 < 27$ (cubes!)

4 $\sqrt{30} \approx 5.5$ $5.4^2 = 29.16$, $5.5^2 = 30.25 \rightarrow$ closer to 30.25

5 FALSE $(-2)^2 = +4$. $\sqrt{(-4)}$ is not a real number

Your turn: 10 minutes

Worksheet pages 4–5 • Section 5 • work on your own • Support Box on page 3

Must

A1–A8 and B1–B4

Should

B5–B8

Could

Section C (C1–C5)

Extension

Section D (optional)

All correct in Round 2? Start at A8, then Section B. Calculator: multiply only — no $\sqrt{\quad}$ or $\sqrt[3]{\quad}$ keys.

Spot the Mistake

Worksheet page 6 · pairs · circle the first error, explain it, correct it

E1 Sam

$$\sqrt{36} = 18 \text{ because } 36 \div 2 = 18$$

E2 Lina

$$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16} = 3 + 4 = 7$$

E3 Omar

$$36 < 40 < 49, \text{ so } 6 < \sqrt{40} < 7. \text{ 40 is between 36 and 49, so } \sqrt{40} \approx 6.5$$

Spot the Mistake: corrections

E1: 6, not 18

Square root is not $\div 2$. Check: $6^2 = 36$ ✓ ($18^2 = 324$)

E2: 5, not 7

Add first: $\sqrt{(9 + 16)} = \sqrt{25} = 5$. You cannot split $\sqrt{\quad}$ over $+$.

E3: 6.3, not 6.5

40 is 4 from 36 but 9 from 49, not halfway. $6.3^2 = 39.69$, $6.4^2 = 40.96 \rightarrow \sqrt{40} \approx 6.3$

Exit ticket: 4 minutes, on your own

Worksheet page 7 • no calculator • hand it in

X1 Calculate (a) $\sqrt{49}$ (b) $\sqrt[3]{27}$

X2 Calculate $\sqrt[3]{-1000}$. Show a check.

X3 (a) Between which two whole numbers is $\sqrt{85}$? (b) Estimate $\sqrt{85}$ to 1 d.p.

X4 A cube has volume 90 cm^3 . Between which two whole numbers is its edge? Explain.

Recap and next step

Today

- $\sqrt{\quad}$ undoes squaring; $\sqrt[3]{\quad}$ undoes cubing
- $\sqrt[3]{\quad}$ of a negative is negative; $\sqrt{\quad}$ of a negative is not real
- Estimate: Trap $\rightarrow$ Roots $\rightarrow$ Closer? $\rightarrow$ Test $\rightarrow$ Decide
- Always check by squaring or cubing

Next step

- After-Lesson Practice: about 25 minutes
- Required: R1, H1–H12, Self-Test
- Short on time: R1, H1, H2, H4, H5, H7, H10, H13, Self-Test
- Optional challenge: H13–H16